MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 02** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer question (2 marks) and 3 Short Answer questions (3 marks each).
4. **Section C** comprises 1 Applied Case Study question of 4 marks with internal choice.
5. Use of calculators is not permitted.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. The first four terms of an AP whose first term is -2 and common difference is -2 are: [1]

(A) $-2, 0, 2, 4$ (B) $-2, 4, -8, 16$ (C) $-2, -4, -6, -8$ (D) $-2, -4, -8, -16$

Q2. The 21st term of the AP whose first two terms are -3 and 4 is: [1]

(A) 17 (B) 137 (C) 143 (D) -143

Q3. If the 2nd term of an AP is 13 and the 5th term is 25, what is its 7th term? [1]

(A) 30 (B) 33 (C) 37 (D) 38

Q4. Which term of the AP $21, 42, 63, 84, \ldots$ is 210? [1]

(A) 9th (B) 10th (C) 11th (D) 12th

Q5. **Assertion (A):** The nth term of the AP $a, a+d, a+2d, \ldots$ is given by $a_n = a + (n-1)d$.
Reason (R): Each term is obtained by adding the common difference d to the preceding term. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Short Answer Questions (11 Marks)

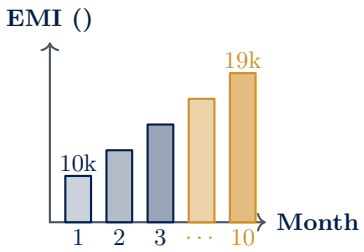
Q6. Which term of the AP $3, 15, 27, 39, \ldots$ will be 120 more than its 21st term? [2]

Q7. Find the sum of all two-digit natural numbers which leave a remainder 1 when divided by 4. [3]

Q8. The angles of a quadrilateral are in AP. If the greatest angle is double the least angle, find all four angles of the quadrilateral. [3]

Q9. Find three numbers in AP whose sum is 21 and the product of the first and third numbers exceeds the second number by 6. [3]

Q10. CASE STUDY: Automobile Bank Loan Repayment Amortization



Rishi purchased a brand-new electric SUV and financed it with an auto-loan from a nationalized bank. To encourage disciplined early repayment, the bank offers a stepped repayment scheme.
Rishi agrees to pay his total loan through monthly installments. He starts with a first installment of $10,000$ and commits to increasing each subsequent monthly installment by $1,000$ over the preceding month until the entire loan is completely discharged.

Based on the above mathematical scenario, answer the following questions:

(i) Write down the first three terms of the AP representing the monthly installments paid by Rishi. **[1]**

(ii) What amount does Rishi pay to the bank in his 10th monthly installment? **[1]**

(iii) What total cumulative amount has Rishi paid to the bank in the first 12 installments (1 full year)? **[2]**

— OR —

If the total loan principal amount to be cleared was $1,18,000$, how many monthly installments will he need to clear the debt? **[2]**

MATHS MANTHAN TUITION CLASSES

★★★ **END OF QUESTION PAPER · ALL THE BEST** ★★★